

IPL SQL ANALYSIS

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Dataset: IPL matches.csv and deliveries.csv

Dataset link: <https://www.kaggle.com/datasets/nowke9/ipldata>

Import the two CSV files as separate SQL tables named matches and deliveries.

First we create a database named 'ipl' and use it for this task. Next step is to create two tables 'matches' and 'deliveries'. We use the same columns present in the actual data set while creating the table.

The screenshot displays a SQL IDE interface with two panels. The top panel shows the execution of two SQL statements: `CREATE database ipl;` and `USE IPL;`. The bottom panel shows the execution of a `CREATE TABLE matches` statement with the following columns: `id` (INT PRIMARY KEY), `season` (INT), `city` (VARCHAR(50)), `date` (VARCHAR(20)), `team1` (VARCHAR(50)), `team2` (VARCHAR(50)), `toss_winner` (VARCHAR(50)), `toss_decision` (VARCHAR(10)), `result` (VARCHAR(20)), `dl_applied` (INT), `winner` (VARCHAR(50)), `win_by_runs` (INT), `win_by_wickets` (INT), `player_of_match` (VARCHAR(50)), `venue` (VARCHAR(100)), `umpire1` (VARCHAR(50)), `umpire2` (VARCHAR(50)), and `umpire3` (VARCHAR(50)).

Top Panel Output:

#	Time	Action	Message
1	12:02:11	Create database ipl	1 row(s) affected
2	12:02:29	USE IPL	0 row(s) affected

Bottom Panel Output:

#	Time	Action	Message
1	12:02:11	Create database ipl	1 row(s) affected
2	12:02:29	USE IPL	0 row(s) affected
3	12:09:12	CREATE TABLE matches (id INT PRIMARY KEY, season INT, city VARCHAR(50), date VARCHAR(20), team1 VARCHAR(50), team2 VARCHAR(50), toss_winner VARCHAR(50), toss_decision VARCHAR(10), result VARCHAR(20), dl_applied INT, winner VARCHAR(50), win_by_runs INT, win_by_wickets INT, player_of_match VARCHAR(50), venue VARCHAR(100), umpire1 VARCHAR(50), umpire2 VARCHAR(50), umpire3 VARCHAR(50));	0 row(s) affected

```

27 • CREATE TABLE deliveries (
28     match_id      INT,
29     inning         INT,
30     batting_team   VARCHAR(50),
31     bowling_team   VARCHAR(50),
32     `over`         INT,
33     ball           INT,
34     batsman        VARCHAR(50),
35     non_striker    VARCHAR(50),
36     bowler         VARCHAR(50),
37     is_super_over  INT,
38     wide_runs      INT,
39     bye_runs       INT,
40     legbye_runs    INT,
41     noball_runs    INT,
42     penalty_runs   INT,
43     batsman_runs   INT,
44     extra_runs     INT,
45     total_runs     INT,
46     player_dismissed VARCHAR(50),
47     dismissal_kind VARCHAR(30),
48     fielder        VARCHAR(50),
49     CONSTRAINT fk_deliveries_match FOREIGN KEY (match_id) REFERENCES matches(id)
50 );

```

Output				
Action Output				
#	Time	Action	Message	
✓ 1	14:26:53	USE IPL	0 row(s) affected	
✓ 2	14:26:57	DROP table deliveries	0 row(s) affected	
✓ 3	14:27:02	CREATE TABLE deliveries (match_id INT, inning INT, batting_team VARCHAR(50), bowling_team VARCHAR(50), `over` ...	0 row(s) affected	

Next step we load the data from CSV to MYSQL

```

56 • LOAD DATA LOCAL INFILE 'D:/Suhair/Data_Analysis_Class/Final_Task/dataset/matches.csv'
57 INTO TABLE matches
58 FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
59 LINES TERMINATED BY '\n'
60 IGNORE 1 ROWS;
61
62 • SELECT * FROM matches;

```

Result Grid														
date	team1	team2	loss_winner	loss_decision	result	d_applied	winner	win_by_runs	win_by_wickets	player_of_match	venue	umpire1	umpire2	
2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field	normal	0	Sunrisers Hyderabad	35	0	Yuvraj Singh	Rajiv Gandhi International Stadium, Uppal	AY Dandekar	NJ Long	
2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field	normal	0	Rising Pune Supergiant	0	7	SPD Smith	Maharashtra Cricket Association Stadium	A Nand Kishore	S Ravi	
2017-04-07	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	field	normal	0	Kolkata Knight Riders	0	10	CA Lynn	Saurashtra Cricket Association Stadium	Nitin Menon	CK Nandan	
2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	field	normal	0	Kings XI Punjab	0	6	GJ Maxwell	Holkar Cricket Stadium	AK Chaudhary	C Shamsuddin	

Output				
Action Output				
#	Time	Action	Message	
✗ 1	12:26:28	LOAD DATA LOCAL INFILE 'D:/Suhair/Data_Analysis_Class/Final_Task/dataset/matches.csv' INTO TABLE matches FIELDS TERMINATED BY ',' OP...	Error Code: 1046. No database selected Select the default DB to be used by double-clicking its name in the SCHEMAS list in the sidebar.	
✓ 2	12:26:34	USE IPL	0 row(s) affected	
✓ 3	12:26:39	LOAD DATA LOCAL INFILE 'D:/Suhair/Data_Analysis_Class/Final_Task/dataset/matches.csv' INTO TABLE matches FIELDS TERMINATED BY ',' OP...	756 row(s) affected Records: 756 Deleted: 0 Skipped: 0 Warnings: 0	
✓ 4	12:26:59	SELECT * FROM matches LIMIT 0, 1000	756 row(s) returned	

```

64 • LOAD DATA LOCAL INFILE 'D:/Suhair/Data_Analysis_Class/Final_Task/dataset/deliveries.csv'
65 INTO TABLE deliveries
66 FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
67 LINES TERMINATED BY '\n'
68 IGNORE 1 ROWS;
69
70 • SELECT * FROM deliveries;

```

match_id	inning	batting_team	bowling_team	over	ball	batsman	non_striker	bowler	is_super_over	wide_runs	bye_runs	legbye_runs	noball_runs	penalty_runs	batsman_runs	extra_runs	total_runs	player_dismissed	dismissal_kind	fielder
5	1	Royal Challengers B...	Delhi Daredevils	6	4	Mandeep ...	SR Watson	PJ Cummins	0	0	0	0	0	0	0	0	0			
5	1	Royal Challengers B...	Delhi Daredevils	6	5	Mandeep ...	SR Watson	PJ Cummins	0	0	0	0	0	0	0	0	0	Mandeep Singh	bowled	
5	1	Royal Challengers B...	Delhi Daredevils	6	6	KM Jadhav	SR Watson	PJ Cummins	0	0	0	0	0	0	0	0	0			
5	1	Royal Challengers B...	Delhi Daredevils	7	1	SR Watson	KM Jadhav	S Nadeem	0	0	0	0	0	0	1	0	1			
5	1	Royal Challengers B...	Delhi Daredevils	7	2	KM Jadhav	SR Watson	S Nadeem	0	0	0	0	0	0	1	0	1			

Output				
Action Output				
#	Time	Action	Message	
✓ 1	14:26:53	USE IPL	0 row(s) affected	
✓ 2	14:26:57	DROP table deliveries	0 row(s) affected	
✓ 3	14:27:02	CREATE TABLE deliveries (match_id INT, inning INT, batting_team VARCHAR(50), bowling_team VARCHAR(50), `over` ...	0 row(s) affected	
✓ 4	14:27:44	LOAD DATA LOCAL INFILE 'D:/Suhair/Data_Analysis_Class/Final_Task/dataset/deliveries.csv' INTO TABLE deliveries FIELDS TERMINATED BY ',' ...	179078 row(s) affected, 1 warning(s): 1261 Row 179078 doesn't contain data for all columns Records: 179078 Deleted: 0 Skipped: 0 Warnings: 1	
✓ 5	14:28:01	SELECT * FROM deliveries LIMIT 0, 1000	1000 row(s) returned	

Part A: Basic Queries and Filtering

1. Display the first 10 records from the matches table.

```
72
73 • SELECT * FROM matches LIMIT 10;
```

	id	season	city	date	team1	team2	toss_winner	toss_decision	result	d_applied	winner	win_by_runs	win_by_wickets	player_of_match	venue	umpire1
▶	1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field	normal	0	Sunrisers Hyderabad	35	0	Yuvraj Singh	Rajiv Gandhi International Stadium, Uppal	AY Dandekar
	2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field	normal	0	Rising Pune Supergiant	0	7	SPD Smith	Maharashtra Cricket Association Stadium	A Nand Kishore
	3	2017	Rajkot	2017-04-07	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	field	normal	0	Kolkata Knight Riders	0	10	CA Lynn	Saurashtra Cricket Association Stadium	Nitin Menon
	4	2017	Indore	2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	field	normal	0	Kings XI Punjab	0	6	GJ Maxwell	Holkar Cricket Stadium	AK Chaudhary
	5	2017	Bangalore	2017-04-08	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore	bat	normal	0	Royal Challengers Bangalore	15	0	KM Jadhav	M Chinnaswamy Stadium	
	6	2017	Hyderabad	2017-04-09	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad	field	normal	0	Sunrisers Hyderabad	0	9	Rashid Khan	Rajiv Gandhi International Stadium, Uppal	A Deshmukh
	7	2017	Mumbai	2017-04-09	Kolkata Knight Riders	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	4	N Rana	Wankhede Stadium	Nitin Menon
	8	2017	Indore	2017-04-10	Royal Challengers Bangalore	Kings XI Punjab	Royal Challengers Bangalore	bat	normal	0	Kings XI Punjab	0	8	AK Patel	Holkar Cricket Stadium	AK Chaudhary
	9	2017	Pune	2017-04-11	Delhi Daredevils	Rising Pune Supergiant	Rising Pune Supergiant	field	normal	0	Delhi Daredevils	97	0	SV Samson	Maharashtra Cricket Association Stadium	AY Dandekar
	10	2017	Mumbai	2017-04-12	Sunrisers Hyderabad	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	4	JJ Bumrah	Wankhede Stadium	Nitin Menon

2. Display the first 10 records from the deliveries table.

```
74
75 • SELECT * FROM deliveries LIMIT 10;
```

	match_id	inning	batting_team	bowling_team	over	ball	batman	non_striker	bowler	is_super_over	wide_runs	bye_runs	legbye_runs	noball_runs	penalty_runs	batman_runs	extra_runs	total_runs	player_dismissed	dismissal_kind	fielder
▶	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	1	DA Warner	S Dhawan	TS Mills	0	0	0	0	0	0	0	0	0			
	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	2	DA Warner	S Dhawan	TS Mills	0	0	0	0	0	0	0	0	0			
	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	3	DA Warner	S Dhawan	TS Mills	0	0	0	0	0	0	0	4	0	4		
	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	4	DA Warner	S Dhawan	TS Mills	0	0	0	0	0	0	0	0	0			
	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	5	DA Warner	S Dhawan	TS Mills	0	2	0	0	0	0	0	0	2	2		
	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	6	S Dhawan	DA Warner	TS Mills	0	0	0	0	0	0	0	0	0			
	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	7	S Dhawan	DA Warner	TS Mills	0	0	0	1	0	0	0	1	1			
	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	2	1	S Dhawan	DA Warner	A Choudhary	0	0	0	0	0	0	0	1	0	1		
	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	2	2	DA Warner	S Dhawan	A Choudhary	0	0	0	0	0	0	0	4	0	4		
	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	2	3	DA Warner	S Dhawan	A Choudhary	0	0	0	0	1	0	0	0	1	1		

3. Count the total number of rows in the matches table.

```
77 • SELECT COUNT(*) AS total_matches FROM matches;
```

```
78
```

	total_matches
▶	756

4. Count the total number of rows in the deliveries table.

```
79 • SELECT COUNT(*) AS total_deliveries FROM deliveries;
```

	total_deliveries
▶	179078

5. Display all unique IPL seasons from the matches table.

```
81 • SELECT DISTINCT season FROM matches ORDER BY season;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
season			
▶ 2008			
2009			
2010			
2011			
2012			
2013			
2014			
2015			
2016			
2017			
2018			
2019			

6. Display all unique cities where IPL matches were played.

```
83
```

```
84 • SELECT DISTINCT city FROM matches;
```

```
85
```

```
86
```

Result Grid	Filter Rows:	Export:	Wrap Cell Cont
city			
▶ Hyderabad			
Pune			
Rajkot			
Indore			
Bangalore			
Mumbai			
Kolkata			
Delhi			
Chandigarh			
Kanpur			
Jaipur			
Chennai			
Cape Town			

7. Display all unique team names appearing in the team1 column.

```
87 • SELECT DISTINCT team1 FROM matches;
88
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	team1
▶	Sunrisers Hyderabad
	Mumbai Indians
	Gujarat Lions
	Rising Pune Supergiant
	Royal Challengers Bangalore
	Kolkata Knight Riders
	Delhi Daredevils
	Kings XI Punjab
	Chennai Super Kings
	Rajasthan Royals
	Deccan Chargers
	Kochi Tuskers Kerala
	Pune Warriors
	Rising Pune Supergiants
	Delhi Capitals

8. Display all matches played in the 2017 season.

```
89
90 • SELECT * FROM matches WHERE season = 2017;
91
```

Result Grid | Filter Rows: | Export/Import: | Wrap Cell Content:

	id	season	city	date	team1	team2	toss_winner	toss_decision	result	d_applied	winner	win_by_runs	win_by_wickets	player_of_match	venue	umpire1
▶	1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field	normal	0	Sunrisers Hyderabad	35	0	Yuvraj Singh	Rajiv Gandhi International Stadium, Uppal	AY Dandekar
	2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field	normal	0	Rising Pune Supergiant	0	7	SPD Smith	Maharashtra Cricket Association Stadium	A Nand Kishore
	3	2017	Rajkot	2017-04-07	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	field	normal	0	Kolkata Knight Riders	0	10	CA Lynn	Saurashtra Cricket Association Stadium	Nitin Menon
	4	2017	Indore	2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	field	normal	0	Kings XI Punjab	0	6	GJ Maxwell	Holkar Cricket Stadium	AK Chaudhary
	5	2017	Bangalore	2017-04-08	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore	bat	normal	0	Royal Challengers Bangalore	15	0	KN Jadhav	M Chinnaswamy Stadium	Nitin Menon
	6	2017	Hyderabad	2017-04-09	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad	field	normal	0	Sunrisers Hyderabad	0	9	Rashid Khan	Rajiv Gandhi International Stadium, Uppal	A Deshmukh
	7	2017	Mumbai	2017-04-09	Kolkata Knight Riders	Mumbai Indians	Mumbai Indians	bat	normal	0	Mumbai Indians	0	4	N Rana	Wankhede Stadium	Nitin Menon
	8	2017	Indore	2017-04-10	Royal Challengers Bangalore	Kings XI Punjab	Royal Challengers Bangalore	bat	normal	0	Kings XI Punjab	0	8	AR Patel	Holkar Cricket Stadium	AK Chaudhary
	9	2017	Pune	2017-04-11	Delhi Daredevils	Rising Pune Supergiant	Rising Pune Supergiant	field	normal	0	Delhi Daredevils	97	0	SV Samson	Maharashtra Cricket Association Stadium	AY Dandekar
	10	2017	Mumbai	2017-04-12	Sunrisers Hyderabad	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	4	JJ Bumrah	Wankhede Stadium	Nitin Menon
	11	2017	Kolkata	2017-04-13	Kings XI Punjab	Kolkata Knight Riders	Kolkata Knight Riders	field	normal	0	Kolkata Knight Riders	0	8	SP Narine	Eden Gardens	A Deshmukh
	12	2017	Bangalore	2017-04-14	Royal Challengers Bangalore	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	4	KA Pollard	M Chinnaswamy Stadium	KN Ananthapadmanabhan
	13	2017	Rajkot	2017-04-14	Rising Pune Supergiant	Gujarat Lions	Gujarat Lions	field	normal	0	Gujarat Lions	0	7	AJ Tye	Saurashtra Cricket Association Stadium	A Nand Kishore
	14	2017	Kolkata	2017-04-15	Kolkata Knight Riders	Sunrisers Hyderabad	Sunrisers Hyderabad	field	normal	0	Kolkata Knight Riders	17	0	RV Uthappa	Eden Gardens	AY Dandekar
	15	2017	Delhi	2017-04-15	Delhi Daredevils	Kings XI Punjab	Delhi Daredevils	bat	normal	0	Delhi Daredevils	51	0	CJ Anderson	Feroz Shah Kotla	VC Barde
	16	2017	Mumbai	2017-04-16	Gujarat Lions	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	6	N Rana	Wankhede Stadium	A Nand Kishore

9. Display all matches where the toss decision was field.

```
92
93 • SELECT * FROM matches WHERE toss_decision = 'field';
94
95
```

Result Grid | Filter Rows: | Export/Import: | Wrap Cell Content:

	id	season	city	date	team1	team2	toss_winner	toss_decision	result	d_applied	winner	win_by_runs	win_by_wickets	player_of_match	venue	umpire1
▶	1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field	normal	0	Sunrisers Hyderabad	35	0	Yuvraj Singh	Rajiv Gandhi International Stadium, Uppal	AY Dandekar
	2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field	normal	0	Rising Pune Supergiant	0	7	SPD Smith	Maharashtra Cricket Association Stadium	A Nand Kishore
	3	2017	Rajkot	2017-04-07	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	field	normal	0	Kolkata Knight Riders	0	10	CA Lynn	Saurashtra Cricket Association Stadium	Nitin Menon
	4	2017	Indore	2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	field	normal	0	Kings XI Punjab	0	6	GJ Maxwell	Holkar Cricket Stadium	AK Chaudhary
	6	2017	Hyderabad	2017-04-09	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad	field	normal	0	Sunrisers Hyderabad	0	9	Rashid Khan	Rajiv Gandhi International Stadium, Uppal	A Deshmukh
	7	2017	Mumbai	2017-04-09	Kolkata Knight Riders	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	4	N Rana	Wankhede Stadium	Nitin Menon
	9	2017	Pune	2017-04-11	Delhi Daredevils	Rising Pune Supergiant	Rising Pune Supergiant	field	normal	0	Delhi Daredevils	97	0	SV Samson	Maharashtra Cricket Association Stadium	AY Dandekar
	10	2017	Mumbai	2017-04-12	Sunrisers Hyderabad	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	4	JJ Bumrah	Wankhede Stadium	Nitin Menon
	11	2017	Kolkata	2017-04-13	Kings XI Punjab	Kolkata Knight Riders	Kolkata Knight Riders	field	normal	0	Kolkata Knight Riders	0	8	SP Narine	Eden Gardens	A Deshmukh
	12	2017	Bangalore	2017-04-14	Royal Challengers Bangalore	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	4	KA Pollard	M Chinnaswamy Stadium	KN Ananthapadmanabhan
	13	2017	Rajkot	2017-04-14	Rising Pune Supergiant	Gujarat Lions	Gujarat Lions	field	normal	0	Gujarat Lions	0	7	AJ Tye	Saurashtra Cricket Association Stadium	A Nand Kishore
	14	2017	Kolkata	2017-04-15	Kolkata Knight Riders	Sunrisers Hyderabad	Sunrisers Hyderabad	field	normal	0	Kolkata Knight Riders	17	0	RV Uthappa	Eden Gardens	AY Dandekar
	16	2017	Mumbai	2017-04-16	Gujarat Lions	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	6	N Rana	Wankhede Stadium	A Nand Kishore
	17	2017	Bangalore	2017-04-16	Rising Pune Supergiant	Royal Challengers Bangalore	Royal Challengers Bangalore	field	normal	0	Rising Pune Supergiant	27	0	BA Stokes	M Chinnaswamy Stadium	KN Ananthapadmanabhan
	19	2017	Hyderabad	2017-04-17	Sunrisers Hyderabad	Kings XI Punjab	Kings XI Punjab	field	normal	0	Sunrisers Hyderabad	5	0	B Kumar	Rajiv Gandhi International Stadium, Uppal	AY Dandekar
	20	2017	Rajkot	2017-04-18	Royal Challengers Bangalore	Gujarat Lions	Gujarat Lions	field	normal	0	Royal Challengers Bangalore	21	0	CH Gayle	Saurashtra Cricket Association Stadium	S Ravi

10. Display all matches won by Mumbai Indians.

96 **SELECT * FROM matches WHERE winner = 'Mumbai Indians';**

97

98

Result Grid

Filter Rows

Edit:     Export/Import:   Wrap Cell Content: 

	id	season	city	date	team1	team2	loss_winner	toss_decision	result	d_applied	winner	win_by_runs	win_by_wickets	player_of_match	venue	umpire1	umpire2
7	1017	Mumbai	2017-04-09	Kolkata Knight Riders	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	4	N Rana	Wankhede Stadium	Nitin Menon	CK Nand	CK Nand
10	1017	Mumbai	2017-04-12	Surreys Hyderabad	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	4	JJ Bumrah	Wankhede Stadium	Nitin Menon	CK Nand	CK Nand
12	1017	Bangalore	2017-04-14	Royal Challengers Bangalore	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	4	KA Pollard	M Chinnayya Stadium	KN Ananthapadmanabhan	AK Chaudh	AK Chaudh
16	1017	Bangalore	2017-04-16	Gujarat Lions	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	6	M Rana	Wankhede Stadium	A Nand Kohore	S Ravi	S Ravi
22	1017	Indore	2017-04-20	Kings XI Punjab	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	8	JC Butler	Holkar Cricket Stadium	M Erasmus	CS Sham	CS Sham
24	1017	Mumbai	2017-04-22	Mumbai Indians	Delhi Daredevils	Delhi Daredevils	field	normal	0	Mumbai Indians	14	0	MI McCleaghnan	Wankhede Stadium	A Nand Kohore	S Ravi	S Ravi
34	1017	Rajkot	2017-04-29	Gujarat Lions	Mumbai Indians	Gujarat Lions	bat	tie	0	Mumbai Indians	0	0	KN Pandya	Saurashtra Cricket Association Stadium	AK Chaudhary	CB Gaffa	CB Gaffa
37	1017	Mumbai	2017-05-01	Royal Challengers Bangalore	Mumbai Indians	Royal Challengers Bangalore	bat	normal	0	Mumbai Indians	0	5	RG Sharma	Wankhede Stadium	AK Chaudhary	CB Gaffa	CB Gaffa
44	1017	Delhi	2017-05-05	Mumbai Indians	Delhi Daredevils	Delhi Daredevils	field	normal	0	Mumbai Indians	146	0	UWP Simmons	Feroz Shah Kotla	Nitin Menon	CK Nand	CK Nand
53	1017	Kolkata	2017-05-13	Mumbai Indians	Kolkata Knight Riders	Kolkata Knight Riders	field	normal	0	Mumbai Indians	9	0	AT Rayudu	Eden Gardens	A Nand Kohore	S Ravi	S Ravi
58	1017	Bangalore	2017-05-19	Kolkata Knight Riders	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	6	KV Sharma	M Chinnayya Stadium	NJ Llong	Nitin Men	Nitin Men
59	1017	Hyderabad	2017-05-19	Mumbai Indians	Rising Pune Supergiant	Mumbai Indians	bat	normal	0	Mumbai Indians	1	0	KN Pandya	Rajiv Gandhi International Stadium, U...	NJ Llong	S Ravi	S Ravi
75	1008	Kolkata	2008-04-29	Kolkata Knight Riders	Mumbai Indians	Kolkata Knight Riders	bat	normal	0	Mumbai Indians	0	7	ST Jayaraj	Eden Gardens	BF Bowden	AV Jayay	AV Jayay
82	1008	Mumbai	2008-05-04	Mumbai Indians	Delhi Daredevils	Delhi Daredevils	field	normal	0	Mumbai Indians	29	0	Dr PM Pollock	Dr P M Patel Sports Academy	S Howell	RE Keert	RE Keert
86	1008	Bangalore	2008-05-07	Rajin Royals	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	7	A Nehra	Dr P M Patel Sports Academy	D NE K	RE Keert	RE Keert
90	1008	Bangalore	2008-05-28	Royal Challengers Bangalore	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	9	CRD Fernando	M Chinnayya Stadium	BF Bowden	AV Jayay	AV Jayay

11. Display matches where `win_by_runs` is greater than 50.

99 100 101

99 • SELECT * FROM matches WHERE win_by_runs > 50;

Result Grid Filter Rows Edit Export/Import Wrap Cell Contents

id	season	city	date	team1	team2	tos_winner	tos_loss	result	d_applied	winner	win_by_runs	win_by_wickets	player_of_match	venue	unpre1
15	2017	Pune	2017-04-11	Delhi Daredevils	Rising Pune Supergiant	Delhi Daredevils	bat	normal	0	Delhi Daredevils	97	0	SI Swaminathan	Maharashtra Cricket Association Stadium	AT Dandekar
20	2017	Delhi	2017-04-11	Delhi Daredevils	Rising Pune Supergiant	Delhi Daredevils	bat	normal	0	Delhi Daredevils	51	0	CJ Anderson	Feroz Shah Kotla	YC Boudie
27	2017	Kolkata	2017-04-23	Kolkata Knight Riders	Royal Challengers Bangalore	Royal Challengers Bangalore	field	normal	0	Kolkata Knight Riders	82	0	NM Coulter-Nile	Eden Gardens	CB Gaffaney
33	2017	Pune	2017-04-29	Rising Pune Supergiant	Royal Challengers Bangalore	Royal Challengers Bangalore	field	normal	0	Rising Pune Supergiant	61	0	LH Ferguson	Maharashtra Cricket Association Stadium	KN Ananthap
44	2017	Delhi	2017-05-06	Mumbai Indians	Delhi Daredevils	Delhi Daredevils	field	normal	0	Mumbai Indians	146	0	LHP Simmons	Feroz Shah Kotla	Nitin Menon
60	2008	Bangalore	2008-04-18	Kolkata Knight Riders	Royal Challengers Bangalore	Royal Challengers Bangalore	field	normal	0	Kolkata Knight Riders	140	0	BB McCullum	M Chinnayyasa Stadium	Asad Rauf
69	2008	Chandigarh	2008-04-25	Kings XI Punjab	Mumbai Indians	Mumbai Indians	field	normal	0	Kings XI Punjab	66	0	KC Sangakkara	Punjab Cricket Association Stadium, Mohali	Alem Der
105	2008	Jaisalmer	2008-05-17	Rajasthan Royals	Royal Challengers Bangalore	Royal Challengers Bangalore	field	normal	0	Rajasthan Royals	65	0	GC Smith	Sawai Mansingh Stadium	BF Bowden
111	2008	Mumbai	2008-05-30	Rajasthan Royals	Delhi Daredevils	Delhi Daredevils	field	normal	0	Rajasthan Royals	105	0	R Watson	Wankhede Stadium	BF Bowden
119	2009	Cape Town	2009-04-18	Royal Challengers Bangalore	Rajasthan Royals	Rajasthan Royals	bat	normal	0	Rajasthan Royals	75	0	R Dravid	Newlands	BR Doctrove
122	2009	Port Elizabeth	2009-04-20	Chennai Super Kings	Royal Challengers Bangalore	Chennai Super Kings	bat	normal	0	Chennai Super Kings	92	0	M Muralitharan	St George's Park	GeJ Berling
132	2009	Port Elizabeth	2009-04-27	Mumbai Indians	Kolkata Knight Riders	Mumbai Indians	bat	normal	0	Mumbai Indians	92	0	SM Tendulkar	St George's Park	GeJ Berling

12. Display matches where win_by_wickets equals 10.

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SELECT * FROM matches WHERE win_by_wickets = 10;

Result Grid

Filter Rows

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13. Display matches where the toss winner and match winner are the same team.

105 • SELECT * FROM matches WHERE toss_winner = winner;

106

107

Result Grid

Filter Rows

Export/Import

Wrap Cell Contents

id	season	city	date	team1	team2	toss_winner	toss_decision	result	d_applied	winner	win_by_runs	win_by_wickets	player_of_match	venue	umpire1
2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field	normal	0	Rising Pune Supergiant	0	7	SPO Smith	Maharashtra Cricket Association Stadium	A Nand Kishore
3	2017	Rajkot	2017-04-07	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	bat	normal	0	Kolkata Knight Riders	0	10	CJ Lynn	Saurashtra Cricket Association Stadium	Nitin Menon
4	2017	Indore	2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	field	normal	0	Kings XI Punjab	0	6	G Maxwell	Holkar Cricket Stadium	AK Chaudhary
5	2017	Bangalore	2017-04-08	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore	bat	normal	0	Royal Challengers Bangalore	15	0	KM Jadhav	M Chinnaswamy Stadium	
6	2017	Hyderabad	2017-04-09	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad	bat	normal	0	Sunrisers Hyderabad	0	9	Rashid Khan	Rajiv Gandhi International Stadium, Uppal	A Deshmukh
7	2017	Mumbai	2017-04-09	Kolkata Knight Riders	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	4	N Rana	Wankhede Stadium	Nitin Menon
10	2017	Mumbai	2017-04-12	Sunrisers Hyderabad	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	4	JJ Bumrah	Wankhede Stadium	Nitin Menon
11	2017	Kolkata	2017-04-13	Kings XI Punjab	Kolkata Knight Riders	Kolkata Knight Riders	bat	normal	0	Kolkata Knight Riders	0	8	SP Harne	Eden Gardens	A Deshmukh
12	2017	Bangalore	2017-04-14	Royal Challengers Bangalore	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	4	KA Pollard	M Chinnaswamy Stadium	KN Anantbhadran
13	2017	Rajkot	2017-04-14	Rising Pune Supergiant	Gujarat Lions	Gujarat Lions	bat	normal	0	Gujarat Lions	0	7	AJ Tye	Saurashtra Cricket Association Stadium	A Nand Kishore
15	2017	Dehi	2017-04-15	Delhi Daredevils	Kings XI Punjab	Delhi Daredevils	bat	normal	0	Delhi Daredevils	51	0	CJ Anderson	Feroz Shah Kotla	YC Bader
16	2017	Mumbai	2017-04-16	Gujarat Lions	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	6	N Rana	Wankhede Stadium	A Nand Kishore

matches 18

14. Display matches where Mumbai Indians appears as either team1 or team2.

107

108 • SELECT * FROM matches WHERE team1 = 'Mumbai Indians' OR team2 = 'Mumbai Indians';

109

Result Grid

Filter Rows:

Export/Imports

Wrap Cell Contents

	id	season	city	date	team1	team2	toss_winner	toss_decision	result	d_applied	winner	win_by_runs	win_by_wickets	player_of_match	venue	umpire1
▶	2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field	normal	0	Rising Pune Supergiant	0	7	SPD Smith	Maharashtra Cricket Association Stadium	A Nand Kishore
	7	2017	Mumbai	2017-04-09	Kolkata Knight Riders	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	4	N Rana	Wankhede Stadium	Nitin Menon
	10	2017	Mumbai	2017-04-12	Sunrisers Hyderabad	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	4	JJ Bumrah	Wankhede Stadium	Nitin Menon
	12	2017	Bangalore	2017-04-14	Royal Challengers Bangalore	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	4	KA Pollard	M Chinnaswamy Stadium	KH Ananthapadmanabhan
	16	2017	Mumbai	2017-04-16	Gujarat Lions	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	6	N Rana	Wankhede Stadium	A Nand Kishore
	22	2017	Indore	2017-04-20	Kings XI Punjab	Mumbai Indians	Mumbai Indians	field	normal	0	Mumbai Indians	0	8	JC Buttler	Holkar Cricket Stadium	M Erasmus
	24	2017	Mumbai	2017-04-22	Mumbai Indians	Dehi Daredevils	Dehi Daredevils	field	normal	0	Mumbai Indians	14	0	KJ McClenaghan	Wankhede Stadium	A Nand Kishore
	28	2017	Mumbai	2017-04-24	Rising Pune Supergiant	Mumbai Indians	Mumbai Indians	field	normal	0	Rising Pune Supergiant	3	0	BA Stokes	Wankhede Stadium	A Nand Kishore
	34	2017	Rajkot	2017-04-29	Gujarat Lions	Mumbai Indians	Gujarat Lions	bat	tie	0	Mumbai Indians	0	0	KH Pandya	Saurashtra Cricket Association Stadium	AK Chaudhary
	37	2017	Mumbai	2017-05-01	Royal Challengers Bangalore	Mumbai Indians	Royal Challengers Bangalore	bat	normal	0	Mumbai Indians	0	5	RG Sharma	Wankhede Stadium	AK Chaudhary
	44	2017	Dehi	2017-05-06	Mumbai Indians	Dehi Daredevils	Dehi Daredevils	field	normal	0	Mumbai Indians	146	0	LMP Simmons	Feroz Shah Kotla	Nitin Menon
	47	2017	Hyderabad	2017-05-08	Mumbai Indians	Sunrisers Hyderabad	Mumbai Indians	bat	normal	0	Sunrisers Hyderabad	0	7	S Dhawan	Raiv Gandhi International Stadium, Udaol	KH Ananthapadmanabhan

matches 18

15. Display the 10 matches with the highest win_by_runs values.

111 • SELECT * FROM matches ORDER BY win_by_runs DESC LIMIT 10; 112

Result Grid

Filter Rows:

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16. Display the 10 matches with the highest win_by_wickets values.

114 • SELECT * FROM matches ORDER BY win_by_wickets DESC LIMIT 10;

115

Result Grid

Filter Rows

115

Part B – Grouping and Aggregation

17. Count the number of matches played in each season.

```
120 • SELECT season, COUNT(*) AS matches_played
121     FROM matches
122     GROUP BY season
123     ORDER BY season;
124
```

Result Grid | Filter Rows: | Export: | Wrap Cell

	season	matches_played
▶	2008	58
	2009	57
	2010	60
	2011	73
	2012	74
	2013	76
	2014	60
	2015	59
	2016	60
	2017	59
	2018	60
	2019	60

18. Count the number of matches played in each city.

```
126 • SELECT city, COUNT(*) AS matches_played
127     FROM matches
128     GROUP BY city
129     ORDER BY matches_played DESC;
130
```

Result Grid | Filter Rows: | Export: | Wrap Cell Cc

	city	matches_played
▶	Mumbai	101
	Kolkata	77
	Delhi	74
	Bangalore	66
	Hyderabad	64
	Chennai	57
	Jaipur	47
	Chandigarh	46
	Pune	38
	Durban	15
	Bengaluru	14
	Visakhapat...	13
	Centurion	12

19. Count the number of matches played at each venue.

```
131
132 • SELECT venue, COUNT(*) AS matches_played
133 FROM matches
134 GROUP BY venue
135 ORDER BY matches_played DESC;
136
```

Result Grid		
	Filter Rows:	Export:  Wrap Cell Content: 
	venue	matches_played
▶	Eden Gardens	77
	M Chinnaswamy Stadium	73
	Wankhede Stadium	73
	Feroz Shah Kotla	67
	Rajiv Gandhi International Stadium, Uppal	56
	MA Chidambaram Stadium, Chepauk	49
	Sawai Mansingh Stadium	47
	Punjab Cricket Association Stadium, Mohali	35
	Maharashtra Cricket Association Stadium	21
	Dr DY Patil Sports Academy	17
	Subrata Roy Sahara Stadium	17
	Kingsmead	15
	Punjab Cricket Association IS Bindra Stadi...	14

Result 24 x

20. Count the number of matches won by each team.

```
138 • SELECT winner, COUNT(*) AS wins
139     FROM matches
140     GROUP BY winner
141     ORDER BY wins DESC;
142
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content

	winner	wins
▶	Mumbai Indians	109
	Chennai Super Kings	100
	Kolkata Knight Riders	92
	Royal Challengers Bangalore	84
	Kings XI Punjab	82
	Rajasthan Royals	75
	Delhi Daredevils	67
	Sunrisers Hyderabad	58
	Deccan Chargers	29
	Gujarat Lions	13
	Pune Warriors	12
	Rising Pune Supergiant	10
	Delhi Capitals	10

Result 25 ×

21. Count the number of times each team won the toss.

```
144 • SELECT toss_winner, COUNT(*) AS toss_wins
145     FROM matches
146     GROUP BY toss_winner
147     ORDER BY toss_wins DESC;
148
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	toss_winner	toss_wins
▶	Mumbai Indians	98
	Kolkata Knight Riders	92
	Chennai Super Kings	89
	Royal Challengers Bangalore	81
	Kings XI Punjab	81
	Delhi Daredevils	80
	Rajasthan Royals	80
	Sunrisers Hyderabad	46
	Deccan Chargers	43
	Pune Warriors	20
	Gujarat Lions	15
	Delhi Capitals	10
	Kochi Tuskers Kerala	8

Result 26 ×

22. Count the number of bat and field toss decisions.

```
149
150 • SELECT toss_decision, COUNT(*) AS total
151 FROM matches
152 GROUP BY toss_decision;
153
154
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	toss_decision	total			
▶	field	463			
	bat	293			

23. Find the top 10 players with the most Player of the Match awards.

```
154
155 • SELECT player_of_match, COUNT(*) AS No_Of_awards
156 FROM matches
157 GROUP BY player_of_match
158 ORDER BY No_Of_awards DESC
159 LIMIT 10;
160
```

Result Grid			Filter Rows:	Export:	Wrap Cell C
	player_of_match	No_Of_awards			
▶	CH Gayle	21			
	AB de Villiers	20			
	RG Sharma	17			
	DA Warner	17			
	MS Dhoni	17			
	YK Pathan	16			
	SR Watson	15			
	SK Raina	14			
	G Gambhir	13			
	MEK Hussey	12			

24. Find the average win_by_runs for each season.

```
161
162 • SELECT season, AVG(win_by_runs) AS avg_win_by_runs
163 FROM matches
164 GROUP BY season
165 ORDER BY season;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content

	season	avg_win_by_runs
▶	2008	12.1552
	2009	13.4035
	2010	16.2667
	2011	15.0411
	2012	12.9730
	2013	16.3289
	2014	10.7333
	2015	14.4068
	2016	11.2667
	2017	13.3559
	2018	11.2500
	2019	11.0833

25. Find the average win_by_wickets for each season.

```
167
168 • SELECT season, AVG(win_by_wickets) AS avg_win_by_wickets
169 FROM matches
170 GROUP BY season
171 ORDER BY season;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	season	avg_win_by_wickets
▶	2008	3.8103
	2009	3.1579
	2010	3.1667
	2011	3.6301
	2012	3.2568
	2013	2.9868
	2014	3.7500
	2015	2.5085
	2016	4.0667
	2017	3.4576
	2018	3.1000
	2019	3.3667

26. Find the maximum win_by_runs recorded in each season.

```
173
174 • SELECT season, MAX(win_by_runs) AS max_win_by_runs
175 FROM matches
176 GROUP BY season
177 ORDER BY season;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	season	max_win_by_runs			
▶	2008	140			
	2009	92			
	2010	98			
	2011	111			
	2012	86			
	2013	130			
	2014	93			
	2015	138			
	2016	144			
	2017	146			
	2018	102			
	2019	118			

27. Find the maximum win_by_wickets recorded in each season.

```
179
180 • SELECT season, MAX(win_by_wickets) AS max_win_by_wickets
181 FROM matches
182 GROUP BY season
183 ORDER BY season;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	season	max_win_by_wickets			
▶	2008	10			
	2009	10			
	2010	10			
	2011	10			
	2012	10			
	2013	10			
	2014	9			
	2015	10			
	2016	10			
	2017	10			
	2018	10			
	2019	9			

Part C - Ball-by-Ball Analysis

28. Count the total number of sixes (batsman_runs = 6) in the deliveries table.

190

191 • `SELECT COUNT(*) AS total_sixes FROM deliveries WHERE batsman_runs = 6;`

192

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	total_sixes			
▶	8170			

29. Count the total number of fours (batsman_runs = 4) in the deliveries table.

193

194 • `SELECT COUNT(*) AS total_fours FROM deliveries WHERE batsman_runs = 4;`

195

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	total_fours			
▶	20392			

30. Find the total number of wide runs in the deliveries table.

196

197 • `SELECT SUM(wide_runs) AS total_wide_runs FROM deliveries;`

198

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	total_wide_runs			
▶	6576			

31. Find the total number of no-ball runs in the deliveries table.

199

```
200 • SELECT SUM(noball_runs) AS total_noball_runs FROM deliveries;
```

201

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	total_noball_runs			
▶	749			

32. Find the total batsman runs scored by each batsman and show the top 10 run scorers.

202

```
203 • SELECT batsman, SUM(batsman_runs) AS total_runs
204 FROM deliveries
205 GROUP BY batsman
206 ORDER BY total_runs DESC
207 LIMIT 10;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	batsman	total_runs			
▶	V Kohli	5434			
	SK Raina	5415			
	RG Sharma	4914			
	DA Warner	4741			
	S Dhawan	4632			
	CH Gayle	4560			
	MS Dhoni	4477			
	RV Uthappa	4446			
	AB de Villiers	4428			
	G Gambhir	4223			

33. Find the number of sixes hit by each batsman and show the top 10 six hitters.

```
209
210 • SELECT batsman, COUNT(*) AS sixes
211 FROM deliveries
212 WHERE batsman_runs = 6
213 GROUP BY batsman
214 ORDER BY sixes DESC
215 LIMIT 10;
216
```

	batsman	sixes
▶	CH Gayle	327
	AB de Villiers	214
	MS Dhoni	207
	SK Raina	195
	RG Sharma	194
	V Kohli	191
	DA Warner	181
	SR Watson	177
	KA Pollard	175
	YK Pathan	161

34. Find the number of fours hit by each batsman and show the top 10 four hitters.

```
217
218 • SELECT batsman, COUNT(*) AS fours
219 FROM deliveries
220 WHERE batsman_runs = 4
221 GROUP BY batsman
222 ORDER BY fours DESC
223 LIMIT 10;
224
```

	batsman	fours
▶	S Dhawan	526
	SK Raina	495
	G Gambhir	492
	V Kohli	482
	DA Warner	459
	RV Uthappa	436
	RG Sharma	431
	AM Rahane	405
	CH Gayle	376
	PA Patel	366

35. Find the total runs scored by each batting team.

```
225
226 • SELECT batting_team, SUM(total_runs) AS total_runs
227 FROM deliveries
228 GROUP BY batting_team
229 ORDER BY total_runs DESC;
230
```

batting_team	total_runs
Mumbai Indians	29809
Royal Challengers Bangalore	28126
Kings XI Punjab	27893
Kolkata Knight Riders	27419
Chennai Super Kings	26418
Delhi Daredevils	24388
Rajasthan Royals	22431
Sunrisers Hyderabad	17059
Deccan Chargers	11463
Pune Warriors	6358
Gujarat Lions	4862
Delhi Capitals	2630
Rising Pune Supergiant	2470
Rising Pune Supergiants	2063
Kochi Tuskers Kerala	1901

36. Find the total extra runs received by each batting team.

```
232 • SELECT batting_team, SUM(extra_runs) AS total_extras
233 FROM deliveries
234 GROUP BY batting_team
235 ORDER BY total_extras DESC;
236
```

batting_team	total_extras
Mumbai Indians	1645
Kolkata Knight Riders	1524
Kings XI Punjab	1425
Royal Challengers Bangalore	1351
Chennai Super Kings	1314
Delhi Daredevils	1273
Rajasthan Royals	1090
Sunrisers Hyderabad	809
Deccan Chargers	578
Pune Warriors	318
Gujarat Lions	233
Kochi Tuskers Kerala	143
Rising Pune Supergiants	101
Rising Pune Supergiant	100
Delhi Capitals	100

37. Count the number of deliveries bowled by each bowler and show the top 10.

```
238 • SELECT bowler, COUNT(*) AS deliveries_bowled
239 FROM deliveries
240 GROUP BY bowler
241 ORDER BY deliveries_bowled DESC
242 LIMIT 10;
243
```

Result Grid			Filter Rows:	Export:	Wrap C
	bowler	deliveries_bowled			
▶	Harbhajan Singh	3451			
	A Mishra	3172			
	PP Chawla	3157			
	R Ashwin	3016			
	SL Malinga	2974			
	DJ Bravo	2711			
	B Kumar	2707			
	P Kumar	2637			
	UT Yadav	2605			
	SP Narine	2600			

38. Find the total runs conceded by each bowler using total_runs and show the top 10 highest totals.

```
245 • SELECT bowler, SUM(total_runs) AS runs_conceded
246 FROM deliveries
247 GROUP BY bowler
248 ORDER BY runs_conceded DESC
249 LIMIT 10;
250
```

Result Grid			Filter Rows:	Export:	Wrap C
	bowler	runs_conceded			
▶	PP Chawla	4153			
	Harbhajan Singh	4050			
	A Mishra	3850			
	DJ Bravo	3733			
	UT Yadav	3640			
	SL Malinga	3511			
	R Ashwin	3391			
	P Kumar	3342			
	B Kumar	3264			
	RA Jadeja	3221			

39. Count the number of recorded dismissals for each bowler, excluding run out, and show the top 10.

```
251
252 • SELECT bowler, COUNT(*) AS dismissals
253 FROM deliveries
254 WHERE player_dismissed IS NOT NULL
255 AND dismissal_kind <> 'run out'
256 GROUP BY bowler
257 ORDER BY dismissals DESC
258 LIMIT 10;
```

259

Result Grid			Filter Rows:	Export:
	bowler	dismissals		
▶	Harbhajan Singh	3440		
	A Mishra	3163		
	PP Chawla	3150		
	R Ashwin	3003		
	SL Malinga	2956		
	B Kumar	2699		
	DJ Bravo	2690		
	P Kumar	2625		
	UT Yadav	2588		
	SP Narine	2585		

40. Count how many dismissals occurred for each dismissal_kind.

```
261 • SELECT dismissal_kind, COUNT(*) AS total
262 FROM deliveries
263 WHERE dismissal_kind IS NOT NULL
264 AND dismissal_kind <> ''
265 GROUP BY dismissal_kind
266 ORDER BY total DESC;
```

Result Grid			Filter Rows:	Export:	Wrap Ce
	dismissal_kind	total			
▶	caught	5348			
	bowled	1581			
	run out	852			
	lbw	540			
	stumped	278			
	caught and bowled	211			
	retired hurt	12			
	hit wicket	10			
	obstructing the field	2			

Part D - Analysis Using Both Tables

41. Join matches and deliveries using matches.id = deliveries.match_id and display 20 joined records.

```
274 • SELECT m.id, m.season, m.venue, m.winner,  
275         d.inning, d.batting_team, d.bowling_team, d.`over`, d.ball,  
276         d.batsman, d.bowler, d.total_runs  
277 FROM matches m  
278 JOIN deliveries d ON m.id = d.match_id  
279 LIMIT 20;  
280
```

Result Grid												
Filter Rows:				Export:		Wrap Cell Content:		Fetch rows:				
	id	season	venue	winner	inning	batting_team	bowling_team	over	ball	batsman	bowler	total_runs
▶	1	2017	Rajiv Gandhi International Stadium, Uppal	Sunrisers Hyderabad	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	1	DA Warner	TS Mills	0
1	1	2017	Rajiv Gandhi International Stadium, Uppal	Sunrisers Hyderabad	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	2	DA Warner	TS Mills	0
1	1	2017	Rajiv Gandhi International Stadium, Uppal	Sunrisers Hyderabad	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	3	DA Warner	TS Mills	4
1	1	2017	Rajiv Gandhi International Stadium, Uppal	Sunrisers Hyderabad	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	4	DA Warner	TS Mills	0
1	1	2017	Rajiv Gandhi International Stadium, Uppal	Sunrisers Hyderabad	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	5	DA Warner	TS Mills	2
1	1	2017	Rajiv Gandhi International Stadium, Uppal	Sunrisers Hyderabad	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	6	S Dhawan	TS Mills	0
1	1	2017	Rajiv Gandhi International Stadium, Uppal	Sunrisers Hyderabad	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	7	S Dhawan	TS Mills	1
1	1	2017	Rajiv Gandhi International Stadium, Uppal	Sunrisers Hyderabad	1	Sunrisers Hyderabad	Royal Challengers Bangalore	2	1	S Dhawan	A Choudhary	1
1	1	2017	Rajiv Gandhi International Stadium, Uppal	Sunrisers Hyderabad	1	Sunrisers Hyderabad	Royal Challengers Bangalore	2	2	DA Warner	A Choudhary	4
1	1	2017	Rajiv Gandhi International Stadium, Uppal	Sunrisers Hyderabad	1	Sunrisers Hyderabad	Royal Challengers Bangalore	2	3	DA Warner	A Choudhary	1
1	1	2017	Rajiv Gandhi International Stadium, Uppal	Sunrisers Hyderabad	1	Sunrisers Hyderabad	Royal Challengers Bangalore	2	4	DA Warner	A Choudhary	6
1	1	2017	Rajiv Gandhi International Stadium, Uppal	Sunrisers Hyderabad	1	Sunrisers Hyderabad	Royal Challengers Bangalore	2	5	DA Warner	A Choudhary	0
1	1	2017	Rajiv Gandhi International Stadium, Uppal	Sunrisers Hyderabad	1	Sunrisers Hyderabad	Royal Challengers Bangalore	2	6	MC Henriq...	A Choudhary	0
1	1	2017	Rajiv Gandhi International Stadium, Uppal	Sunrisers Hyderabad	1	Sunrisers Hyderabad	Royal Challengers Bangalore	2	7	MC Henriq...	A Choudhary	4
1	1	2017	Rajiv Gandhi International Stadium, Uppal	Sunrisers Hyderabad	1	Sunrisers Hyderabad	Royal Challengers Bangalore	3	1	S Dhawan	TS Mills	1
1	1	2017	Rajiv Gandhi International Stadium, Uppal	Sunrisers Hyderabad	1	Sunrisers Hyderabad	Royal Challengers Bangalore	3	2	MC Henriq...	TS Mills	0
1	1	2017	Rajiv Gandhi International Stadium, Uppal	Sunrisers Hyderabad	1	Sunrisers Hyderabad	Royal Challengers Bangalore	3	3	MC Henriq...	TS Mills	0

42. Using the joined tables, find the total runs scored in each IPL season.

```
281  
282 • SELECT m.season, SUM(d.total_runs) AS total_runs  
283 FROM matches m  
284 JOIN deliveries d ON m.id = d.match_id  
285 GROUP BY m.season  
286 ORDER BY m.season;  
287
```

Result Grid		
Filter Rows:		
Export: Wrap Cell Content: ☐		
	season	total_runs
▶	2008	17937
	2009	16353
	2010	18883
	2011	21154
	2012	22453
	2013	22602
	2014	18931
	2015	18353
	2016	18862
	2017	18786
	2018	20706
	2019	20270

43. Using the joined tables, find the total number of sixes in each season.

288

```
289 • SELECT m.season, COUNT(*) AS sixes
290 FROM matches m
291 JOIN deliveries d ON m.id = d.match_id
292 WHERE d.batsman_runs = 6
293 GROUP BY m.season
294 ORDER BY m.season;
295
```

Result Grid	Filter Rows:	Export:	W
season	sixes		
2008	623		
2009	508		
2010	587		
2011	639		
2012	733		
2013	681		
2014	715		
2015	692		
2016	639		
2017	706		
2018	869		
2019	778		

44. Using the joined tables, find the total number of fours in each season.

296

```
297 • SELECT m.season, COUNT(*) AS fours
298 FROM matches m
299 JOIN deliveries d ON m.id = d.match_id
300 WHERE d.batsman_runs = 4
301 GROUP BY m.season
302 ORDER BY m.season;
303
```

Result Grid	Filter Rows:	Export:	Wrap C
season	fours		
2008	1703		
2009	1321		
2010	1709		
2011	1916		
2012	1911		
2013	2054		
2014	1563		
2015	1611		
2016	1633		
2017	1612		
2018	1674		
2019	1685		

45. Using the joined tables, find the total extra runs in each season.

304

```
305 • SELECT m.season, SUM(d.extra_runs) AS total_extras
306 FROM matches m
307 JOIN deliveries d ON m.id = d.match_id
308 GROUP BY m.season
309 ORDER BY m.season;
310
```

Result Grid			Filter Rows:	Export:	Wrap Cell Conte
	season	total_extras			
▶	2008	1128			
	2009	977			
	2010	1129			
	2011	1226			
	2012	1131			
	2013	1115			
	2014	988			
	2015	926			
	2016	899			
	2017	866			
	2018	800			
	2019	819			

46. Using the joined tables, find total runs scored by each batting team in each season.

311

```
312 • SELECT m.season, d.batting_team, SUM(d.total_runs) AS total_runs
313 FROM matches m
314 JOIN deliveries d ON m.id = d.match_id
315 GROUP BY m.season, d.batting_team
316 ORDER BY m.season DESC;
```

Result Grid				Filter Rows:	Export:	Wrap Cell Content:
	season	batting_team	total_runs			
▶	2019	Chennai Super Kings	2599			
	2019	Delhi Capitals	2630			
	2019	Kings XI Punjab	2530			
	2019	Kolkata Knight Riders	2591			
	2019	Mumbai Indians	2783			
	2019	Rajasthan Royals	2285			
	2019	Royal Challengers Bangalore	2287			
	2019	Sunrisers Hyderabad	2565			
	2018	Chennai Super Kings	2920			
	2018	Delhi Daredevils	2435			
	2018	Kings XI Punjab	2295			
	2018	Kolkata Knight Riders	2863			
	2018	Mumbai Indians	2505			
	2018	Rajasthan Royals	2443			
	2018	Royal Challengers Bangalore	2403			
	2018	Sunrisers Hyderabad	2842			
	2017	Delhi Daredevils	2219			
	2017	Gujarat Lions	2412			
	2017	Kings XI Punjab	2207			
	2017	Kolkata Knight Riders	2484			

47. Using the joined tables, find the top 10 run scorers in the 2017 season.

```

318
319 • SELECT d.batsman, SUM(d.batsman_runs) AS runs
320 FROM matches m
321 JOIN deliveries d ON m.id = d.match_id
322 WHERE m.season = 2017
323 GROUP BY d.batsman
324 ORDER BY runs DESC
325 LIMIT 10;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Cor
	batsman	runs		
▶	DA Warner	641		
	G Gambhir	498		
	S Dhawan	479		
	SPD Smith	472		
	SK Raina	442		
	HM Amla	420		
	MK Pandey	396		
	KA Pollard	395		
	PA Patel	395		
	RA Tripathi	391		

48. Using the joined tables, find the top 10 venues by total runs scored.

```

327
328 • SELECT m.venue, SUM(d.total_runs) AS total_runs
329 FROM matches m
330 JOIN deliveries d ON m.id = d.match_id
331 GROUP BY m.venue
332 ORDER BY total_runs DESC
333 LIMIT 10;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Cor
	venue	total_runs		
▶	Eden Gardens	23857		
	Wankhede Stadium	23649		
	M Chinnaswamy Stadium	23267		
	Feroz Shah Kotla	20896		
	Rajiv Gandhi International Stadium, Uppal	17093		
	MA Chidambaram Stadium, Chepauk	15696		
	Sawai Mansingh Stadium	14440		
	Punjab Cricket Association Stadium, Mohali	10987		
	Maharashtra Cricket Association Stadium	6871		
	Dr DY Patil Sports Academy	4810		

49. Using the joined tables, find the number of recorded dismissals in each season.

```

335
336 • SELECT m.season, COUNT(d.player_dismissed) AS dismissals
337 FROM matches m
338 JOIN deliveries d ON m.id = d.match_id
339 GROUP BY m.season
340 ORDER BY m.season;

```

season	dismissals
2008	13489
2009	13606
2010	14498
2011	17013
2012	17767
2013	18177
2014	14300
2015	13652
2016	14096
2017	13862
2018	14290
2019	14328

50. Using the joined tables, show each match ID, season, winner, and total runs scored in that match, ordered from highest match total to lowest.

```

343 • SELECT m.id AS match_id, m.season, m.winner, SUM(d.total_runs) AS match_total_runs
344 FROM matches m
345 JOIN deliveries d ON m.id = d.match_id
346 GROUP BY m.id, m.season, m.winner
347 ORDER BY match_total_runs DESC;
348

```

match_id	season	winner	match_total_runs
7937	2018	Kolkata Knight Riders	471
206	2010	Chennai Super Kings	469
50	2017	Kings XI Punjab	453
61	2008	Chennai Super Kings	447
11338	2019	Kolkata Knight Riders	445
11326	2019	Royal Challengers Bangalore	438
7904	2018	Rajasthan Royals	435
259	2011	Delhi Daredevils	433
68	2008	Rajasthan Royals	431
562	2015	Royal Challengers Bangalore	431
7917	2018	Chennai Super Kings	431
7944	2018	Royal Challengers Bangalore	431
11153	2019	Kolkata Knight Riders	429
516	2014	Kings XI Punjab	428
7923	2018	Chennai Super Kings	424
41	2017	Delhi Daredevils	422
176	2010	Mumbai Indians	420
7898	2018	Chennai Super Kings	419
11142	2019	Kolkata Knight Riders	419
486	2014	Kings XI Punjab	418

